AMISR in Africa Workshop

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Report

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Executive Summary

There is compelling scientific need for an incoherent scatter radar (ISR) in Africa to understand the global dynamics of the equatorial ionosphere. In combination with the long-running Jicamarca ISR, global studies of this region can be conducted for the first time. Many recent satellite observational studies have shown significant longitudinal differences in the structure and dynamics of the equatorial ionosphere – with the African sector having the most frequent and severe ionospheric irregularities. These irregularities give rise to significant space weather effects for HF radio propagation, satellite communication and navigation systems. In the last five years, significant investment by the international community has increased the number of small instruments (such as GPS receivers and magnetometers) in Africa that can significantly leverage the scientific return of an ISR by giving regional characterization of the geomagnetic field, electrojet currents, and ionospheric structure. In addition, the creation of the African Union's Pan-African University's Space Science Center – that is helping to coordinate research and educational development in space sciences in association with the Square Kilometer Array, provides tremendous scientific and educational collaborative opportunities across Africa. Because of the compelling scientific and practical space weather needs and the timeliness of opportunities within Ethiopia and Africa, many space scientists within the NSF CEDAR and GEM communities have begun the process of planning for a Advanced Modular Incoherent Scatter Radar (AMISR) to be deployed in Africa in the next five years. This report describes the scientific and scientific rational for such a system and describes the planning efforts of the community to date.

1. **Introduction**

This report is a summary of a three day workshop held 1-3 March 2012 to address the concept of an Advanced Modular Incoherent Scatter Radar (AMISR) in Ethiopia. This workshop was funded by the National Science Foundation with generous support from the Boston College, the host institution. There were 48 people attended the workshop including people from seven countries in Africa, three in Europe and one in South America as shown in *Figure 1*. The purpose of this workshop was to identify the scientific and societal motivations for hosting AMISR in Ethiopia and to address logistical issues associated with this plan, and outline the major hurdles to be overcome. This document summarizes the findings of this meeting.

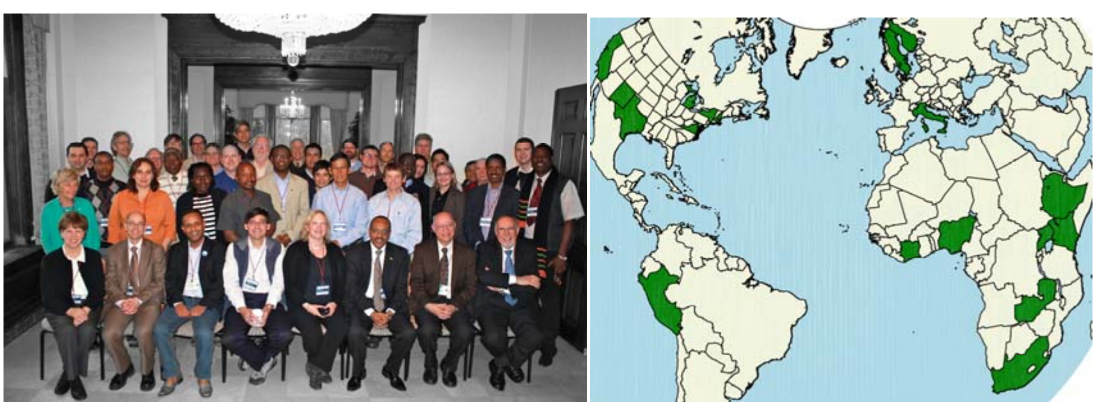

Figure 1. *The workshop participants shown in the left and the country they came from shown in the right side.*

2. Background Introduction – scientific motivations

Geomagnetic Field

NSF and Geospace science strategies and themes include at their core the focus on understanding and unraveling the complex global dynamics of the entire geosphere. The ionosphere is a central region of the geosphere, and within the ionosphere, the equatorial region is one of the most complex, host to numerous instabilities, interactions, and still unresolved questions regarding dynamics. Among the ionospheric phenomena associated with the equator are the ionospheric equatorial anomalies, equatorial spread F, dynamo efficiency, the equatorial electrojet, the strength of the pre-reversal enhancement, and the tidal behavior of thermospheric winds and tides. All of these phenomena are in some way influenced by the regional geomagnetic field, its declination and the proximity of the magnetic to the geographic equator. Jicamarca, Peru (11.95°S, 76.87°W geographic and 0.62°N, 354.6°E) and Bahir Dar, Ethiopia (11.6°N, 37.38°E geographic and 3.0°N, 109.0°E geomagnetic) are in the vicinity of geomagnetic equatorial region. However, the declination of the magnetic field and the magnitude of the magnetic field differ considerably between these two sites. While Jicamarca is located inside the equatorial electrojet region, Bahir Dar is situated barely outside the electrojet region, which is believed to be within the ±2.5° geomagnetic latitudes. By comparing and contrasting measurements, the dependence of many of these phenomena on the magnetic field strength and declination can be studied. An incoherent scatter facility in Ethiopia, in addition to the one already existing in Peru, would enable the study of "microscale to global scale physics", global connections, hemispherical differences, and longitudinal and latitudinal differences.

Space Weather

Space weather is described by the set of conditions on the Sun, and in the solar wind, magnetosphere, ionosphere and thermosphere, which can affect the performance and reliability of space- and ground-based technological systems and can imperil human life. Space weather can also impact, sometimes severely, communication and navigation systems. Several studies have shown clear evidence of space weather-induced adverse effects [*e.g., Makela et al.*, 2001; *Jakowski et al.*, 2005; *Basu et al.*, 2008]. In particular, aviation transportation and satellites are affected due to possible navigation/telecommunication problems. Occasionally space weather impacts can cause severe economic loss [e.g., satellite outages]. With the future advancement of technology the impact of space weather will certainly increase unless suitable protective measures will be taken in advance. Understanding the physics behind each space weather impact and improving the current standard of forecasting space weather impact is a major objective of the space science community. For example, understanding the propagation of waves through the inner-magnetosphere, a region that is very important for communication and navigation satellites, requires knowledge of the density structure of the ionosphere [*Doherty et al.*, 2004]. The ionosphere is the major source of plasma that can significantly influence the propagation of radio waves. Thus, ionospheric disturbances can cause range errors, rapid phase and amplitude fluctuations (radio scintillations) of satellite signals that may lead to degradation of the system performance, its accuracy and reliability.

Despite the fact that much progress has been made in the study of ionospheric density structure and dynamics in the last decade, there are many gaps remaining in our global understanding of the fundamental electrodynamics that governs the formation of equatorial ionospheric density irregularities. The uneven distribution of ground-based instruments has been one of the main barriers to obtaining global understanding of the dynamics and structures of the ionosphere. In regions like Africa, observations of the ionosphere are currently not possible due to the simple lack of ground-based instruments. The lack of ground-based space physics instrumentation (radars, magnetometers, ionosondes, GPS dual frequency receivers, etc.) infrastructure in Africa is easily apparent as shown in *Figure 2*. On the other hand, satellite observations demonstrate that there are large differences in the formation of ionospheric irregularities over Africa as compared to other longitudinal sectors. Small-to-medium scale ionospheric irregularities, such as equatorial plasma bubbles, equatorial spread F (ESF), and bottom side spread F (BSSF) at low/mid-latitudes in the African region have not been studied in detail due to lack of instrumentation. This has created a gap in the global understanding of the physics behind the evolution and formation of plasma irregularities in the equatorial region, which in turn, imposes severe limitations on ionospheric and plasmaspheric density modeling efforts.

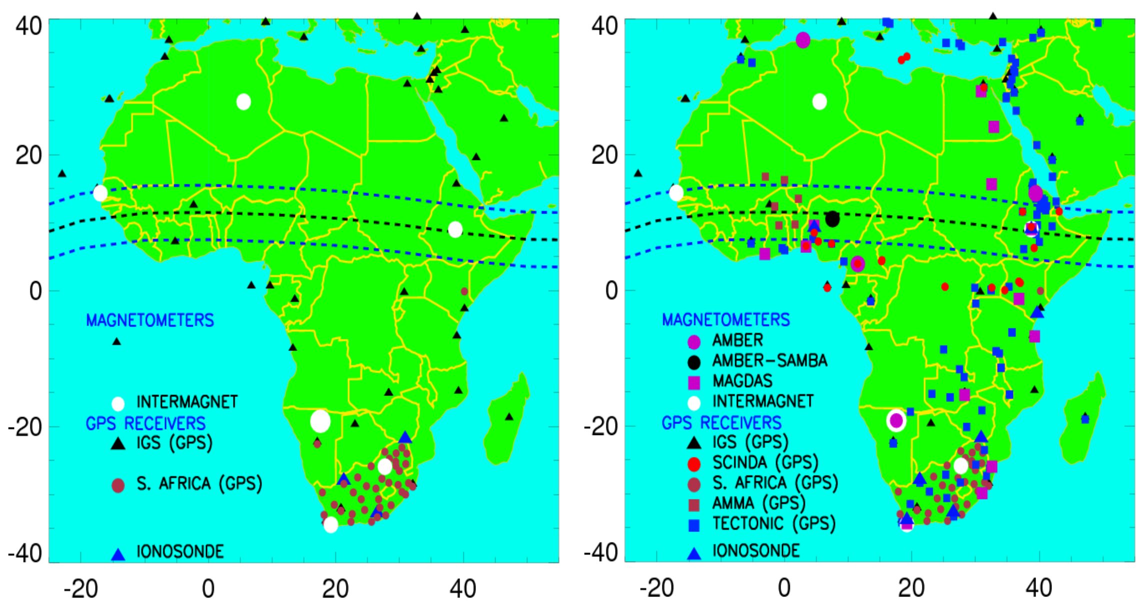

Figure 2. *(left panel) Ground-based instrument coverage in African five years ago, and (right panel) shows the current ground-based instrumentation in Africa.*

Without ground-based instrumentation, the ionospheric density structure in the African sector has traditionally been estimated by model interpolation over vast geographic areas. This makes it very difficult to observe small-to-medium scale ionospheric irregularities at low/mid-latitudes in the region. In fact, our communication and navigation technologies depend on understanding, modeling, and mitigating the effects of these irregularities [*Doherty et al.*, 2004]. For example, when scintillation (the rapid amplitude and phase fluctuations of radio signals due to turbulence generated by ionospheric irregularities) [*Groves et al.*, 1997] occurs the following technological systems will be affected: (1) regional Satellite Communications (SATCOM) outages for extended periods (hours), (2) increased Global Positioning Satellite (GPS) navigation errors, (3) degraded High Frequency (HF) radio communication. At tropical latitudes, there is more danger of satellite signals or communication navigation systems being disrupted [*Kintner et al.*, 2009]. Africa has the most tropical landmass area in the world. In fact, there is more danger of satellite signals or communication navigation systems being disrupted at tropical latitudes [*Kintner et al.*, 2009] than elsewhere. Africa has the most tropical landmass area in the world. *Figure 3* shows where scintillation will most frequently impact Global Navigation Satellite Systems (GNSS) signals. Therefore, understanding the physics behind the equatorial ionospheric irregularities is becoming critical to our daily life.

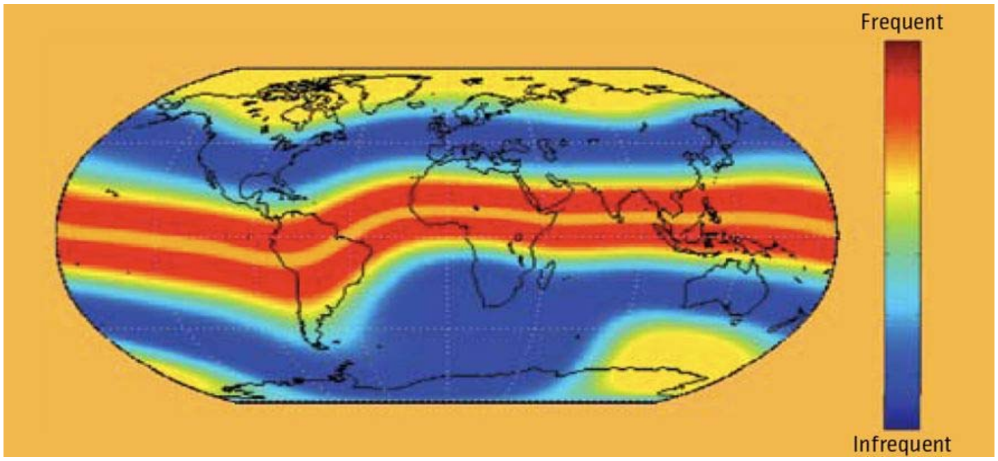

Figure 3: Scintillation map showing the frequency of disturbances at solar maximum. Scintillation is most intense and most frequent in two bands surrounding the magnetic equator, up to 100 days per year. At poleward latitudes, it is less frequent and it is least frequent at mid-latitude, a few to ten days per year.

Lack of knowledge of African ionosphere

Africa has the largest landmass beneath the geomagnetic equator and it serves as a natural platform to study ionospheric disturbances that affect navigation and communication systems. Satellite data have shown that the equatorial ionospheric density structures, especially at the equatorial region in the African continent, respond to space weather effects differently than other parts of the Earth [*Amory-Mazaudier et al.*, 2005]. For example, satellite observations (see *Figure 3*) show unique equatorial ionospheric structures only in the African equatorial sector [*Hei et al.*, 2005; *Su*, 2005; *Burke et al.*, 2006; *Yizengaw et al.*, 2010]. In the African region, ionospheric depleted density irregularities known as bubbles are very active year round unlike other regions and when they form, these bubbles are much deeper and occur more frequently than those observed in other longitudinal sectors [*Hei et al.*, 2005 and *Su*, 2005]. Ionospheric depletions in the African region also rise to high altitudes (up to 1000+ km) more often than those in other longitude sectors [*Burke et al.*, 2004]. New data from satellites (e.g., C/NOFS) have also indicated that the equatorial ionosphere in the African sector responds differently than other sectors. However, none of these differences have been confirmed, validated, or studied in detail by observations from the ground due to the lack of suitable ground-based instrumentation in the region. The source of these unique density irregularities in the continent remains a mystery for the scientific community, becoming a challenging problem for our navigation and communication systems. This causes incomplete global understanding of the physics behind the redistribution of plasma and the evolution and formation of plasma irregularities in the equatorial region, which is the challenging problem for ionospheric and plasmaspheric density modeling efforts. Coordinated ground- and space-based observations of ionospheric irregularities and scintillation activities are needed to unravel the African sector equatorial electrodynamics. Coordinating ISR observations with satellite over flights would be an important next step.

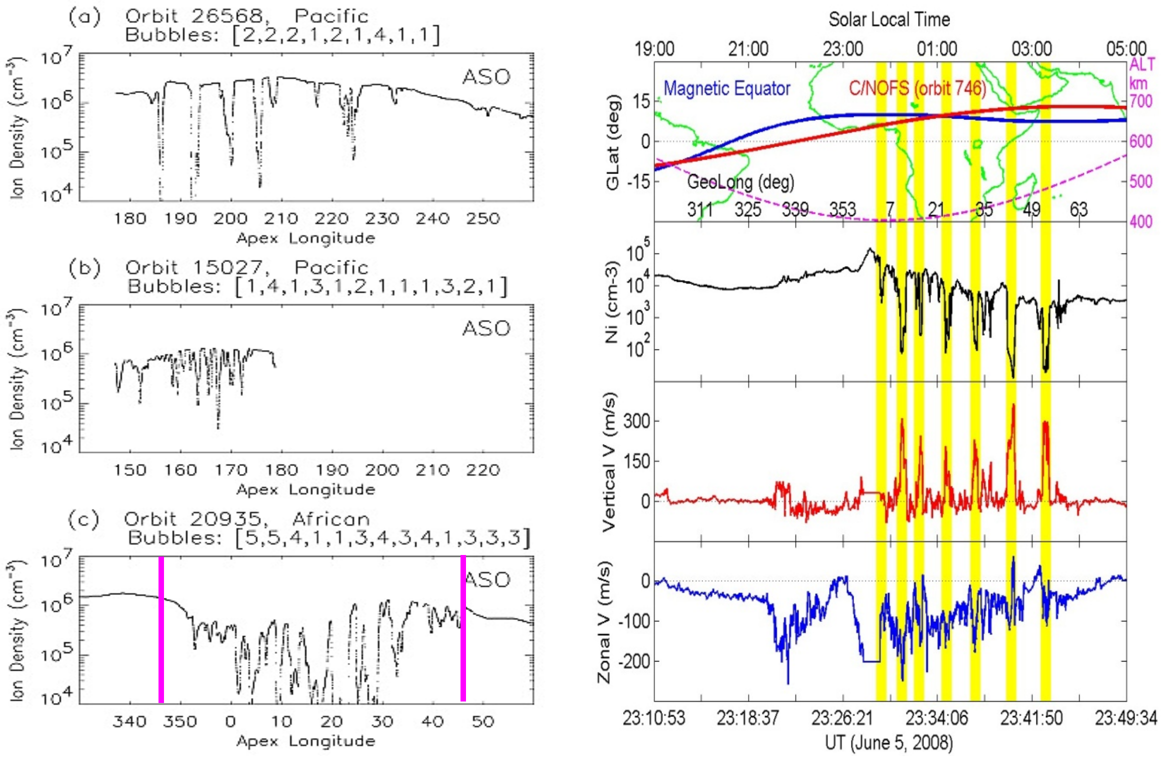

Figure 4. *(Left panels) the typical bubbles observed by AEE satellite that orbits at an approximate altitude of 400 km, occurring in the fall (August, September, October (ASO)): (a) and (b) in the Pacific sector and (c) in the African sector [after Hei et al., 2005]. (Right panels) shows the periodic bubble observation by recently launched C/NOFS satellite. The panels from top to bottom depict the ground track of the satellite (red curve) along with the geomagnetic equator (blue curve), in situ density that shows periodic bubbles that happened only in Africa, the corresponding vertical drift velocity, and zonal velocity, respectively [after de La Beaujardiere et al., 2012].*

3. Why do we need AMISR in Africa?

ISRs, with their measurements of the full profile of electron and ion density, can play a significant role in the development of reliable, global ionospheric models that can be used for forecasting and nowcasting. ISR's deduce height- and time-resolved plasma drift velocities, electron and ion temperatures, electron densities, ion composition, and ion-neutral collision frequencies. These parameters provide further information about the neutral gas, neutral temperatures and winds, and electric fields present in the medium. No other instrument provides all of this ionospheric information. An ISR in Africa would greatly enhance the physics output of other space science instrumentation and would provide information that would greatly enhance our understanding of equatorial electrodynamics.

The African sector has severely lacked ground-based instrumentation for space science (see *Figure 2*). During the past couple of years very few (compared to the land-mass that Africa covers) small instruments, like GPS receivers, magnetometers, ionosondes, and VLF have been deployed in the region. Adding a single incoherent scatter radar (ISR) to the current space science instrumentation in the equatorial African sector would be of significant scientific benefit. An ISR in Africa would greatly enhance the physics output of these other instruments and would provide information that would greatly enhance our understanding of equatorial electrodynamics.

To date, the vast majority of equatorial ISR measurements have been collected at Jicamarca, so that much of what we know about equatorial physics is based on Jicamarca ISR observations. However, Jicamarca is in the American sector where the geomagnetic equator dips, and there is a fairly large excursion between the geomagnetic and geodetic equator. This is shown clearly in Figure 2 of *Kintner et al.* [2009]. On the other hand, in the African sector the geomagnetic equator is fairly well aligned with the geodetic equator. Therefore, having ISR in the African sector will play a vital role in the effort of global understanding of the dynamics and structure of the ionospheric density irregularities, which is the prime candidate for the failure of our navigation and communication systems especially during magnetically active periods.

2.1. Scientific problems to be addressed

2.1.1. Why are ionospheric irregularities in general unique in the African sector? Satellite observations have shown very unique equatorial ionospheric density structures in the African region. *Hei et al*. [2005] has revealed a longitudinal variation in the large-scale bubble properties (zonal width, depletion level, and spacing). *Hei et al*. [2005] found the African region to be the longitude sector where the peak in bubble activity is strongest, containing a predominance of long-wavelength (~1000 km) structures populated with sub-structures of ~100 km-scale within them as shown in left panel of *Figure 4*. No other region in the globe showed similar characteristics. Most recent *in situ* density observations from C/NOFS also reveal similar unique bubble activities. *Figure 4* on the right panel shows periodic bubbles that are separated by ~1000 km (8.1^{o}) in longitude which is almost equidistant between each bubble. The corresponding upward ion velocity (measured from VEFI) reaches as high as 300 m/s inside the bubble region. It is possible that these periodic bubbles are triggered by waves that may be present in the region. However, the dearth of ground-based instrumentation in the region makes it impossible to confirm these unique equatorial ionospheric structures from the ground and that leads the investigation of the physics into speculative dead ends. The unique satellite *in situ* observations have initiated several open questions, which include: *what is special in Africa that it produces such unique ionospheric irregularities? Is it because of special seeding conditions? Or because of special electrodynamics in the region?*

In order to answer these questions and understand the evolution and formation of these unique bubble structures, ground-based observations are required. Thus, placing AMISR in Africa will play a vital role to understand the physics behind the unique equatorial ionospheric structures in the African sector. The combination of ground- and space-based continuous observations will give us a better understanding of ionospheric irregularities as a function local time, season, and magnetic activity.

2.1.2. Why does the depleted plasma (or bubbles) penetrate to higher altitudes in the African sector? It has been reported that the DMSP satellite at the altitude of 940 km observed stronger bubbles in the African sector than those observed at other longitudes [*Gentile et al.*, 2011]. This observation may (see *Figure 5*) indicate that the bubbles penetrate to higher altitudes in the African sector. These bubbles are generated by the equatorial spread F (ESF) phenomenon.

ESF consists of a turbulent gravitational overturn of the F-region in which densities and electric fields cause the development of irregular structures with sizes ranging from hundreds of kilometers down to sub-meter sizes. The coherent scatter mode of ISR instrument is the only suitable instrument that can routinely image the vertical expansion of the bubbles. Therefore, placing AMISR in the African sector is important to understand the unique equatorial ionospheric phenomenon in the region. The AMISR observation provides not only the vertical expansion of the bubbles but also its evolution through time. Both of these are important to understand the seeding mechanisms for the formation of the very deep and wide bubbles that have been often observed by satellites. Currently, the Jicamarca radar routinely observes vertically elongated density bubbles or "plumes" connecting the unstable bottomside to the more stable topside ionosphere [*e.g., Hysell et al.,* 2009 and the references therein]. The classical Rayleigh-Taylor instability (RTI) mechanism has been pointed out as the possible triggering mechanism for the upward extension of the bubbles [*Woodman and LaHoz,* 1976], and the initiation is favored when the F-layer has risen to higher altitudes [*Farley et al.,* 1970; *Jayachandran et al*., 1993; *Sastri et al*., 1997]. Several authors have investigated the role of RTI for the onset and evolution of the ESF phenomena [*Basu and Basu,* 1985; *Mendillo et al.,* 1992; *Jayachandran et al.,* 1993]. *Sultan et al.* [1996] have shown that during storm condition the electric field ($\mathbf{E}\times\mathbf{B}$) plays a strong role in the initiation of RTI and thus in the development of plasma bubbles and its upward extension. However, during magnetically quiet conditions plasma density redistribution plays important role in controlling the onset of ESF. Although promising progress has been made in understanding the connection between ESF, RTI, and bubble extension to higher altitudes, several key questions are still unresolved, especially in the African sector. These are: *Do plasma bubbles (plumes) reach different altitudes in different longitudinal sectors and if so, why? Does the altitude penetration of the bubbles relate to the entire ionospheric plasma uplift or only to the difference between the bubbles' vertically upward motion at different longitudinal sectors? How often and under what conditions does this kind of bubble penetration difference happen? What controls the lifetime of the plasma bubbles? What is the role of gravity waves in influencing the upward extension of the bubbles?*

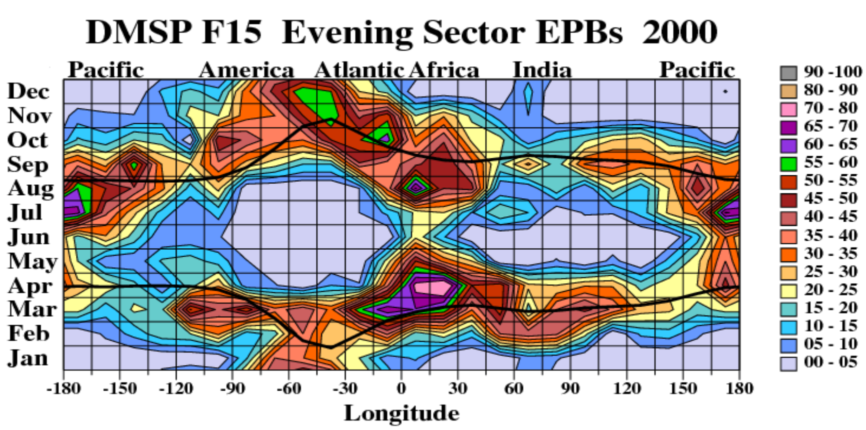

Figure 5. Climatological contour maps of equatorial plasma bubble occurrence rates measured by DMSP satellites between 1989 and 2002 plotted as a function of month versus longitude grid. The solid lines indicate the two times per year when the angle between equatorial declination and the dusk terminator was zero at given longitudes [after Gentile et al., 2011].

The proposed AMISR in Africa will be the only instrument suitable and capable of investigating the dynamics of the thermosphere and ionosphere in the region, enabling us to answer the questions outlined above. Thus, by combining the AMISR in Africa observation with the already available ISR observation at different longitudes, like Jicamarca ISR, the scientific community will be able to closes the gaps of global understanding of the physics behind each event, which will improve the current navigation and communication accuracy.

2.1.3. Why is the ESF phenomenon and thus equatorial bubbles much deeper and more active ***throughout the year*** *in the African region as compared to other longitude sectors?* Although the formation of the bubbles observed over Africa (see *Figure 4*) show unique structure, it can be argued that this is a single pass and may not be indicative of the uniqueness of the African ionosphere. However, the statistical studies of space-based observations (*see Figures 5 and 6*) address this. The observations in these figures show that the density depletions (bubbles) are not only much deeper, as shown in *Figure 4*, but are also more active throughout the year in the African sector compared to other longitudes. Similarly, from C/NOFS observations the statistical bubble occurrence probability, shown in *Figure 7*, also shows that the bubbles in the African sector is active throughout the year compared to other longitudinal sectors. More recently, COSMOS ground observations of M. Yamamoto show a similar longitudinal difference in the variation of the scintillation S_4 index. The COSMOS observations show that S4 is consistently greater over Africa than Asia.

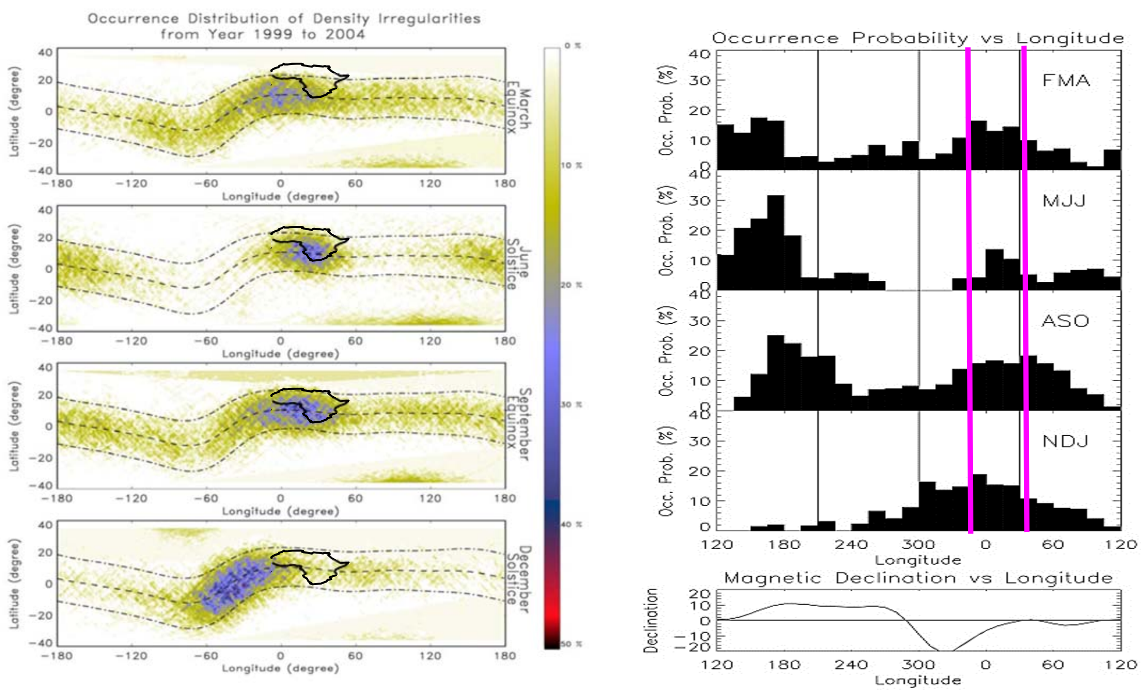

Figure 6. Satellite data suggest both the spatial structure and seasonal behavior of disturbances above Africa are unique. (left panels) seasonal behavior of the density irregularity occurrence probability using ROCSAT in situ density observation [after Su, 2005]. (right panels) bubble occurrence probability as a function of longitude, and the vertical black lines delineate the four longitude sectors (Asian, Pacific, African, and Indian). Magnetic declination in degrees is plotted in the bottom panel as a function of longitude [after Hei et al., 2005]. The pink vertical lines indicate the African region.

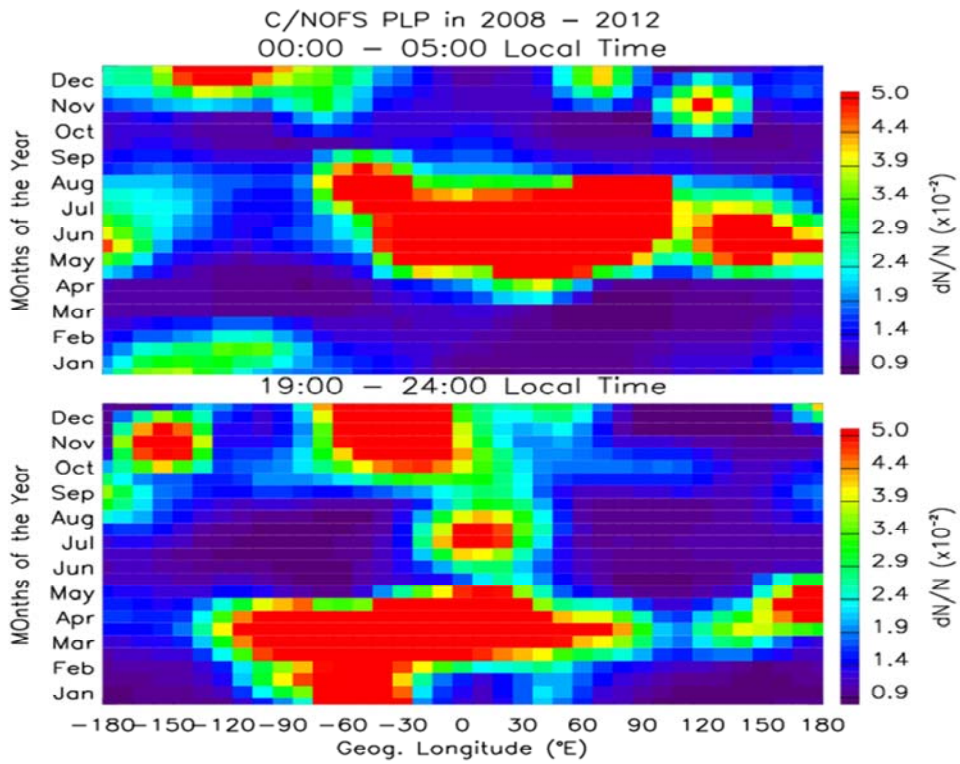

Figure 6. The climatological maps of plasma bubble occurrence probability during dawn (top panel) and dusk (bottom panel) sectors observed by C/NOFS satellites between 2008 and 2012 plotted as a function of month versus longitude grid. All observations were taken when the satellite was below 500 km.

2.1.4. What are the possible governing mechanisms that create unique equatorial structures in Africa? Plasma bubble formation is a complex process in which several factors play key roles for its initiation. It is widely believed that an initial seed perturbation on the bottomside of the ionosphere is required to trigger plasma bubbles. This seed perturbation can be favorable for irregularity development in the case of several ionospheric conditions, which include an elevated F-layer, a steep bottomside density gradient, and a sunset terminator aligned closely with the local geomagnetic meridian [*Scannapieco and Ossakow*, 1976; *Huang and Kelley*, 1996; *Tsunoda*, 1985; *Sultan*, 1996]. The question is what triggers the seed perturbation? Different authors have proposed different possible mechanisms for seed development. These include gravity waves [*Singh et al.*, 1997], a penetration electric field [*Yeh et al.*, 2001], a large-scale wave structure [*Tsunoda*, 2005] and the post-sunset vortex [*Kudeki and Bhattacharyya*, 1999]. It is well understood that the main driving mechanism of the upward extension of the bubble is a vertical drift triggered by the zonal electric field. Because of the geomagnetic field configuration at low-latitudes, the effects of the electric field and vertical $\mathbf{E}\times\mathbf{B}$ drifts strongly influence ionospheric structure [*e.g., Chandra and Rastogi*, 1974; *Anderson et al.*, 2002]. The electric field in turn originates from the ionospheric dynamo electric field and/or solar wind dynamo electric field that can penetrate all the way down to the ionosphere. The Rayleigh-Taylor instability (RTI) also creates intense polarization electric fields within the perturbations, generating $\mathbf{E}\times\mathbf{B}$

drift that cause the bubbles to be uplifted to higher altitudes [*Woodman and LaHoz,* 1976; *Scannapieco and Ossakow,* 1976]. *Kudeki et al.* [2007] has recently suggested that an eastward thermospheric wind is able to drive Pedersen currents and polarize F-region structures by creating westward tilted wave-fronts. Another mechanism was presented by *Tsunoda* [2005, 2006], who suggested that a large-scale polarization electric field can be generated by a sporadic E (E_s) layer instability. He indicated that an electric field could be produced by a Hall polarization current due to an altitude-modulated E_s layer in conjunction with the presence of a wind gradient (or shear) in altitude. *Tsunoda* [2005, 2006] showed that the polarization electric field formed at the E-layer by this instability is mapped to the bottomside of the F-layer, triggering plasma bubbles. The growth rate of this instability was found to be consistently larger (a factor of 10) than the growth rate of the RTI. However, due to the lack of instruments in the region, it has been impossible to measure the night side vertical drift velocity which is believed to be the triggering mechanism of the stronger and deeper bubbles often observed in the African sector by satellites. Therefore, placing AMISR in the African sector will provide both the time evolution of the bubbles as well as its triggering mechanisms (the vertical drift velocity) simultaneously. AMISR can provide estimates of both day and night side vertical and zonal drift velocities. However to extend the AMISR observations, the dayside vertical drift can also be estimated using a pair of magnetometers, which are located at different longitudes, including the meridian where AMISR will be located, in the African sector. This technique how to estimate the vertical drift velocity from the magnetometer observations has been described by [*Anderson et al.,* 2002 & 2004; *Yizengaw et al.* 2011 & 2012]. The first result from the African sector shown in *Figure 6* depicts the typical example of indirectly estimated $\mathbf{E}\times\mathbf{B}$ drift (blue curve in the bottom panel) from a pair of AMBER magnetometer (green curve in the top panel) and INTERMAGNET (red curve in the top panel) data. The pink dots represent the vertical drift estimated by VEFI instrument onboard C/NOFS satellite, showing very good agreement with the drift velocity estimated using magnetometer observations.

2.1.5. Why do existing models fail to detect such unique structures in the African sector? Although it has been shown to be much more accurate than TEC derived from climatological models, such as the International Reference Ionosphere (IRI) [*Bilitza,* 2001] and the Bent model [*Llewellyn and Bent,* 1973], due to the uneven distribution of GPS receivers, the GPS data-based global TEC maps fail to describe the global structure and dynamics of the inner-magnetosphere [*e.g., Mannucci et al.,* 1998]. Even the comparatively more accurate JPL Global Ionosphere Map (GIM) [*Mannucci et al.,* 1998] is not able to track the ionospheric irregularities and gradients during geomagnetic storm activity when the ionosphere is highly disturbed, especially in areas where GPS data coverage is poor, such as Africa. Thus, estimation of TEC in this region must rely on spatial/temporal interpolation or extrapolation [*Ho et al.,* 1997]. Data-driven interpolation in GIM cannot overcome poor data coverage.

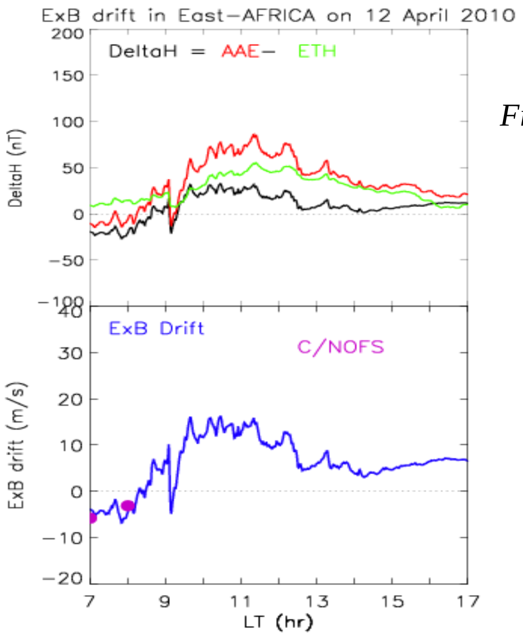

Figure 6. (top panle) the red, green and black curves represent H-component magnetic field at the equator (Addis Ababa), at off the equator at AMBER station (Adigrat), and the difference between red and green curves (ΔH *), respectively. (Bottom panel) and* $\mathbf{E\times B}$ *drift estimated from the* ΔH *curve (blue curve) and pink dots show the ion vertical drift velocity recorded by VEFI onboard C/NOFS satellite (courtesy to R. Pfaff for VEFI data).*

2.2. Potential new science can be address by AMISR in Africa which is not possible with Jicamarca

Unlike Jicamarca radar which is located totally inside the electrojet region because of the position right at the geomagnetic equator, AMISR in Africa, which will be located just outside the electrojet region (~ +3.0°N geomagnetic), will provide a great opportunity to explore the equatorial E-region density structure as well as E-region wind profile for the first time. One reason is the presence of the equatorial electrojet and its associated plasma structuring can typically generate strong coherent echoes and obscure the tenuous incoherent scatter radar signals.

What is the role of the equatorial and/or off-equatorial E-region to balance the current system during the PRE?

The PRE is one of the driving forces for the generation of equatorial ionospheric irregularities, its precise estimation is very important to forecast the ionospheric irregularities and bubbles. Thus, in order to have a precise the PRE estimation, precise measurements of the E-region density, electric fields, and currents are required. *Titheridge* [2000] have shown that even the nighttime E-region densities exhibit appreciable variability in latitude, since starlight production increases in the southern hemisphere. The night time E region measurements in a single magnetic meridian are important for the determination of the evening plasma vortex flow [*Eccles*, 1998] and the steepness of the shear in the zonal plasma flow [*Haerendel et al.*, 1992]. Recent studies show that the post sunset shear in the zonal plasma flow may influence the seeding of equatorial spread F plumes [*Hysell and Kudeki*, 2004]. Therefore, placing AMISR in Africa, which will be barely outside the electrojet region, will play a vital role in estimating the E-region density, current, and electric field. These also allow us to answer other potential questions, like *How does the E-region current system vary when a zonal wind gradient is present? Crain et al.* [1993] have demonstrated that a positive longitudinal (local time) gradient or a west-to-east shear of the F-region zonal wind could also enhance the PRE field.

Another significant importance of E-region density estimation is its contribution for the initiation of ESF, which can be short circuited when the E-region density is substantially large. In spite of the decisive role that the E-region plays during ESF initiation, its contribution to the field line integrated conductivity has never been measured. It has also been argued [*Stephan et al.*, 2002] that 'Sporadic E' (a capricious consequence of wind-shear concentration of meteoric ions) may influence the depleted plasma plume development. Therefore, to study all these electrodynamics effects in the low-latitude ionosphere one must have a precise specification of the electron density and the resulting conductivity of the E-region. Thus, AMISR in Bahir Dar, Ethiopia will be the first capable instrument to address all these.

4. AMISR in Africa Societal Benefits

Placing the AMISR in Africa will attract scientists worldwide and rapidly position Africa at the top of global ionospheric research, creating strong international collaborations, recognition and opportunities for the African Space Science Community. AMISR in Africa has also direct impact in advancing space science research into Africa by establishing and furthering sustainable research/training infrastructure within Africa so that more young scientists will be educated in their own country. It will also play a vital role in the future socioeconomic development of Africa. It will spark interest into the young African generation and encourage them to do science and technology, which is the back bone for the economic development of any country.

A new initiative of the African Union is called the Pan African University (http://www.pau.edu.ng/). The goal of this initiative is to develop a culture of research and graduate education within African Universities. It is developing five centers of excellence at different host universities, each focused on different areas, that brings together 10 additional universities. One of the centers is focused on Space Science and will be located within South Africa. With the awarding of most of the Square Kilometer Array to South Africa, significant investment in radio astronomy will be made. The AMISR in Africa initiative dovetails nicely with this initiative and will allow for new collaborations in terms of radio science research and education.

5. Why Ethiopia is chosen as home for AMISR in Africa?

5.1. ***Its geographic location***: Bahir Dar is located approximately the same geomagnetic latitude as Jicamarca but the two locations has different excursions between the geomagnetic and geographic equator. In addition, Bahir Dar is located nearly outside the electrojet region (11.6°N, 37.38°E geographic and 3.0°N, 109.0°E geomagnetic). Thus, placing AMISR in Bahir Dar will provide great opportunities to the scientific community for new science investigations, including those potential science questions (*see section 3*) which could not be possible to do it using Jicamarca radar just because of it geographic locations.

5.2. ***Operational sustainability***: Operating AMISR sustainably requires well trained local personnel and very good support from local universities and government. In the case of Ethiopia, as it is presented in detail in *section 5* below, the government and host university already have shown enthusiastic logistic and financial support to make this project a reality. Moreover, Bahir Dar University has already ISR specialized scientist, Dr Baylie Damtie. As it is evident in his CV, shown in the appendix, Dr Damtie, who is the current president of Bahir Dar University, has outstanding experience in ISR technique. This provides an assurance for the sustainability of the

AMISR in Africa. Moreover, Bahir Dar University has very diligent record in maintaining the equipments that has been housed in the university compound.

5.3. ***Space Science research activity***: Because of the outstanding public outreach program performed by International Heliospheric Years (IHY) program, many African countries were inspired to become part of the space science community. Some countries even introduced space science program into the curriculum of their universities. Ethiopia is one those countries that came to the forefront in advancing space science activities in the country. Bahir Dar University is one of the two top universities that already launched space science program. With the enthusiastic support of the government, Bahir Dar University quickly became the center of gravity of space science research in the country. The university has already postgraduate program (Masters and PhD program) in space science program.

5.4. ***Accessibility:*** Since Ethiopia is the seat for African Union and many other UN offices, and thus many major airlines have direct flights from major European and North American cities. Therefore, placing AMISR in Ethiopia make it an easily accessible major research facility in the African continent. More importantly, Ethiopia is also an easily accessible country for many African countries, making AMISR an easily accessible major research facility not only for Ethiopian but also for the entire African countries.

6. What Bahir Dar University already offered for the success of AMISR project

Bahir Dar University in particular and Ethiopian government in general expressed their enthusiastic support for the success of this project. As it has been announced at the workshop by the Ethiopian Ambassador in the United State and the president of Bahir Dar University, the following expenses will be covered by the Bahir Dar University. These are: custom fee to import the equipments, land for the site will be provided for free, frames that are used to place the panels will be constructed locally with local expenses, utilities (water, power, internet) shall be provided, security will be provided by the university, and all other necessary logistical support and technical maintenance. On top these, Bahir Dar University will provide a matchup funds for antenna construction cost (*see letter attached*).

AMISR in Africa Workshop Agenda

The prime objective of the workshop is to discuss and identify the scientific and societal benefits of developing a new international upper atmospheric facility, Advanced Modular Incoherent Scatter Radar (AMISR), in Ethiopia. The key points which need to be addressed during the workshop are:

1. Is there any scientific need for placing AMISR observatory in Africa in general and in Ethiopia in particular? Demonstrating the wealth of scientific opportunities that such a facility would provide is the principal goal of this workshop.
2. Are there any feasible technical and logistical issues that need to be addressed to satisfy the scientific need and successful deployment of AMISR so it will augment the already in place global ground-based instruments, such as ionosonde, GPS receivers, magnetometers, etc.
3. Is there any feasible international collaboration model that could lead a project to develop the facility?

It is important that the focus of the workshop is not about what the community has done, but what we would like to do with this facility, which will be placed right at the geomagnetic equator making it the second equatorial ISR facility in the world. The workshop will also demonstrate how placing the AMISR in Africa will advance space science research into Africa by establishing and furthering sustainable research/training infrastructure within Africa.

With these goals in mind the proposed agenda is as follows:

Day 1 – Thursday 1st March 2012

Time	Speaker	Title
Introduction and Scientific capability of AMISR Convener: E. Yizengaw		
08:00-08:45		Registration
08:45-08:55	E. Yizengaw	Introduction and opening comments
08:55-09:10	P. Doherty	Welcome opening remark
09:10-09:25	H.E. Ambassador Girma Birru (Ethiopian Ambassador to the US)	Opening remark, showing the government support of the project
09:25-09:40	B. Robinson	NSF support on AMISR in Africa
09:40-10:05	B. Damtie	Detail briefing about the already available facility at Bahir Dar to host AMISR
10:05-10:30	Coffee break	
Introductory presentation about the capabilities of current Incoherent Scatter Radars and a description of AMISR. Convener: E. Yizengaw		
10:30-11:00	J. Chau	Brief introductory of JIC and its scientific achievements
11:00-11:30	M. Kelley	The historic scientific achievement with ISR instrument at the equator
11:30-12:00	D. Hysell	Scientific importance of AMISR in Africa
12:00-12:30		Open discussion
12:30-13:30	Lunch	
Convener: A. Stromme		
13:30-13:45	J. LaBelle	Potential science experiments that can be done by AMISR in Ethiopia
13:45-14:00	J. Fentzke	Potential Meteor/Dusty Plasma Studies in the African Sector
14:00-14:30	J. Foster	Haystack ISR capability and its scientific achievements
14:30-15:00		Open discussion
15:00-15:30	Coffee break	

On Uniqueness of the African Sector and what AMISR data will provide to understand the physics Convener: A. Stromme		
15:30-16:00	K. Groves/R. Caton	Why Africa is unique from SCINDA observation point of view?
16:00-16:30	O. de la Beaujardiere	C/NOFS's unique observation over Africa
16:30-17:00	M. Moldwin/E. Zesta	Ground- and space-based magnetometer observation over Africa
17:00-18:00		Open discussion
18:30-21:00		Conference dinner reception

Day 2 – Friday 2nd March 2012

Ground-based Space science Instruments in Africa Convener: A. Coster		
08:30-09:00	J. Davila	Past, Present, and Future ground-based instrumentation in Africa
09:00-09:30	L. McKinnell	Past, Present, and Future space science research in Africa
09:30-09:45	Parris/Caton/Groves	A VHF Coherent Backscatter Radar in Africa
09:45-10:00		Open discussion
10:00-10:30	Coffee break	
Technology Convener: M. Moldwin		
10:30-11:00	C. Heinselman	Current AMISR technology
11:00-11:30	F. Lind	Low Frequency Array Technology and The Geospace Array
11:30-11:45	R. Tsunoda	AMISR alternative site proposal
11:45-12:00	M. Kelley	A Proposal to Build a Solar Village in Ethiopia
12:00-12:30		Open discussion
12:30-13:30	Lunch	
13:30-14:00	A. Stromme	About Antarctica ISR technology
14:00-14:30	E. Turunen	EISCAT 3D technology
14:30-15:00		Open discussion
15:00-15:30	Coffee break	
15:30-16:00	J. Sliker/A. Weaver	Solar power options

Scientific and societal opportunities of AMISR for African universities Convener: M. Moldwin and A. Coster		
16:00-16:15	B. Rabiu	Scientific and societal opportunities of AMISR for African universities
16:15-16:30	G. Mengistu	Recent Progress in Geospace Research in Ethiopia from ground and space based instrumentation: Implication for Scientific and societal opportunities of AMISR
16:30-16:45	P. Baki	Scientific and societal opportunities of AMISR for Kenyan universities
16:45-17:00	O. Obrou	AMISR in Africa, an opportunity for space science research: case of the University of Cocody
17:00-17:15	P. Sibanda	TBD
17:15-17:30	J. Adeniyi	TBD
17:30-18:00		Open discussion

Day 3 – Saturday 3rd March 2012

Panel discussion about programmatic and strategy of placing AMISR in Africa, geared towards outlining the key points of the workshop to draft a report Convener: P. Doherty		
08:30-10:00		Identify authors for different sections of the report
		Identify possible funding sources for the construction of the radar
10:00-10:30	Coffee break	Identify possible cheapest way to construct the radar
10:30-11:30		Developing African University and African Union support for a user-facility in Africa

AMISR in Africa Workshop Participants

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		Organizers	
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